

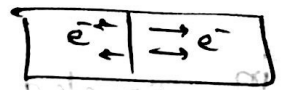
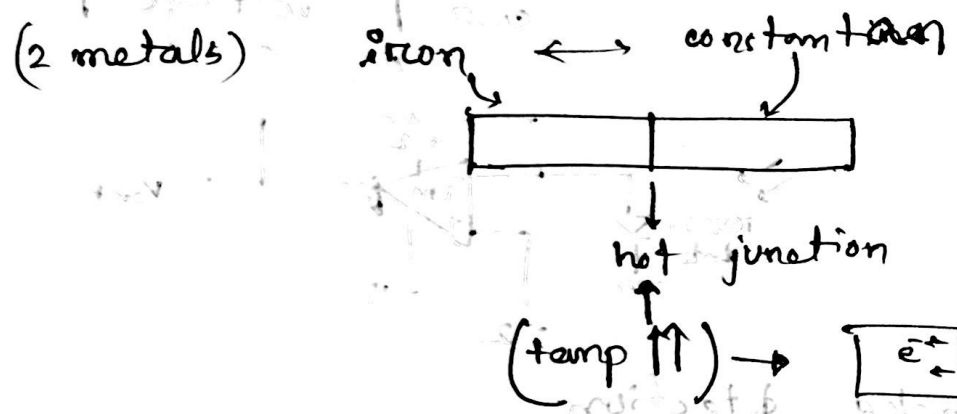
→ PN Junc → heavily doped.

→ reverse current is basis → (photodiode is not).

Temp Sensors :

- * Temp sensitive voltage srcs → LM 35
- * " " current → (1°C ↑ - 10 mV ↑)
- ↳ নিজের পাঠ্য (self study).

• Thermocouples :



as metal diff so e^- flow 2 side
a diff way

Seebeck effect → small voltage produced so



v difference দেখা যায়

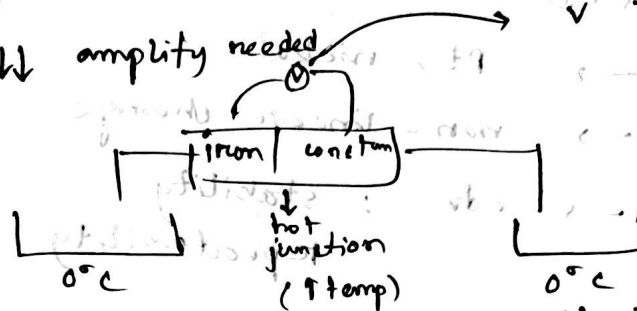
voltage depends on type of metal
→ temp.

V increases 1 to 75 $\mu V/^\circ C$ V $\propto$ temp $\uparrow$

iron, constantan
platinum, alloy of Pt, Rh.

3 main problem

- 1) O/P $\downarrow$ amplity needed
- 2)



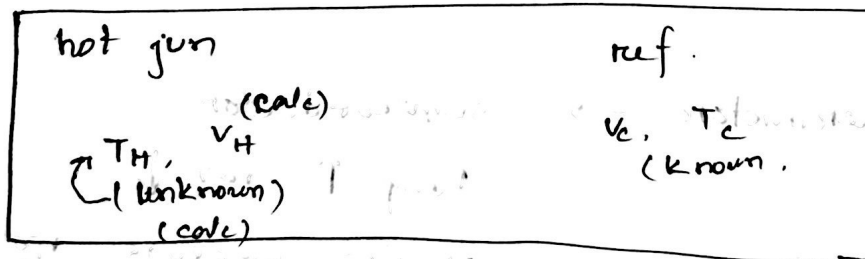
(let)
diff
 $22^\circ C$
So, Temp
change
22°C

old junction
(known temp $\rightarrow$ reference temp.)
(ref. temp.)

change 22°C temp change
possible na

$$V_{ref} \pm \text{temp} = V_s$$

#



$$V = V_H - V_C$$

$$= a(T_H - T_C)$$

seebach effect.

$$V_{diff} = 3.41 \text{ mV}$$

$$T_C = 22^\circ C$$

$$\begin{matrix} 0 \\ 10 \\ \rightarrow 20 \end{matrix}$$

(from table)

$$0.879 = V_C$$

$$3.41 \times 10^{-3} = V_H - 0.879$$

$$\Rightarrow V_H = 4.289 \text{ mV}$$

4.2 & 4.3 μV $\rightarrow$ $104^\circ / 105^\circ$
100°C

Let S_0 , $T_H = 104^\circ\text{C}$

$$3.41 \times 10^{-3} = a(104 - 22)$$

$$\Rightarrow a = \frac{3.41 \times 10^{-3}}{82}$$

RTD

- Pt, nickel
- non-linear change
- adv: stability, repeatability
- $(25^\circ\text{C} \rightarrow 100^\circ\text{C})$ constant change
- -250 to 850°
- temp $\uparrow$ resistance $\uparrow$ voltage drop $\uparrow$
- change in res. $\rightarrow$ converted to Voltage.
- used in exact/precise temp measurement.

Thermistor $\rightarrow$

semiconductor
 temp $\uparrow$ res $\downarrow$
 precise measure
 not applicable
 in expensive

length $\propto$ resistance $\propto$ voltage

Force & Pressure Sensors :

- Strain Gauges :

length $\uparrow$ resistance $\uparrow$
semiconductor material

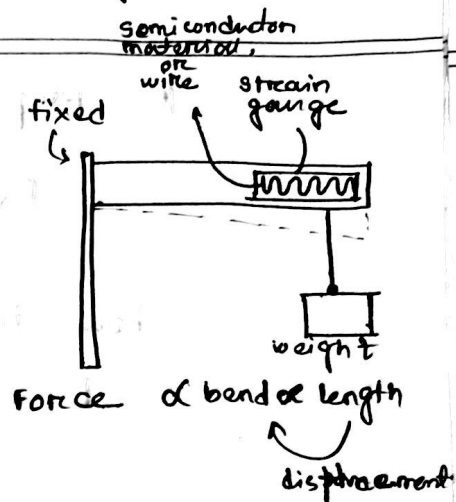
$$V = IR$$

Force $\propto$ length $\propto$ resistance $\propto$ voltage

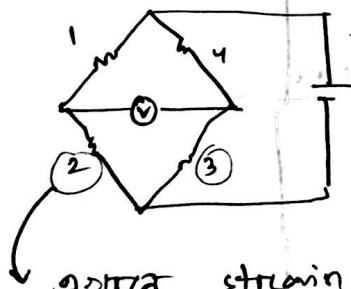
* sensor $\rightarrow$ wheatstone bridge



resistance change ΔR ΔL semiconductor



$$\frac{R_2}{R_4} = \frac{R_1}{R_3} \rightarrow V = 0 \text{ (voltage diff 0)}$$



$$V = \left(\frac{R_3}{R_3 + R_4} - \frac{R_2}{R_1 + R_2} \right) \times V_{ex}$$

अतः strain gauge नागला

as load is applied, V changes. ($R_1, \dots, R_4$ are also changing)

gauge factor $\rightarrow k = \frac{\Delta R/R}{\Delta L/L}$

New resistance $\rightarrow R_{new} = R (1 + k \epsilon)$
 $\epsilon = \Delta L / L$

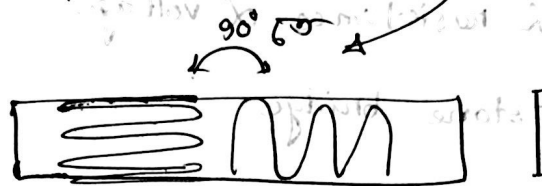
If Temp changes

extra resistance for temp!

$$R_{new} = R(1 + k\varepsilon + \alpha \Delta T)$$

how to nullify $R \propto \Delta T$

→ 2 strain gauge 90° angle



after bending:

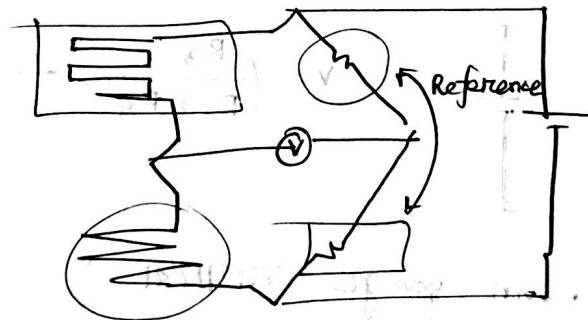


$$R(1 + k\varepsilon)$$



$$R(1 - k\varepsilon)$$

(res ↓)



$$\frac{0}{0} = \frac{\square}{\square}$$

thermistor, RTD

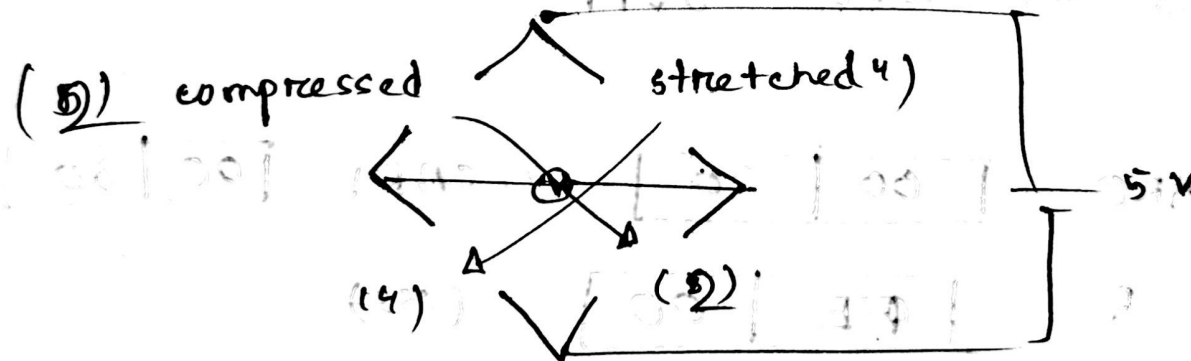
thermocouple 3 problems, working
prc.

Math

stretch (length $\uparrow$) 4Ω

compressed (length $\downarrow$) 2Ω

$$V_{ex} = V_{in} = 5V$$



$$V = \left(\frac{4}{4+5} - \frac{2}{2+4} \right) 5$$

() $\rightarrow$ self study

Analog Interfacing & Industrial Control (Part 1).

Embedded system :

অনেকগুলো (h/w) $\rightarrow$ to do a special task.

$\hookrightarrow$ sensor, μ -controller, actuator.

(s/w) $\rightarrow$ to write code

RTOS : Real Time Operating system

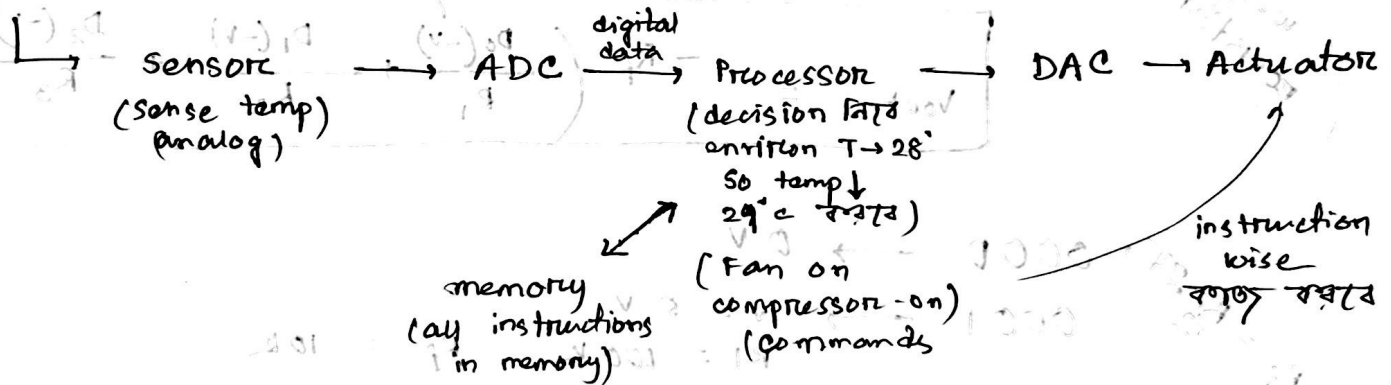
$\nearrow$ different

computer নিচে schedule করে printer, keyboard এর মধ্যে

বসানো যা schedule করে

I/O interface

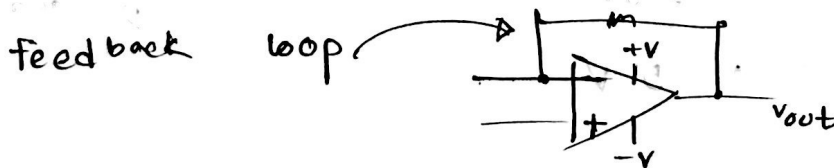
air conditioner, oven, lift $\leftarrow$ Examples.



D/A converter :

binary bit pattern $\rightarrow$ corresponding current/voltage $\rightarrow$ convert.

+ Non inverting point



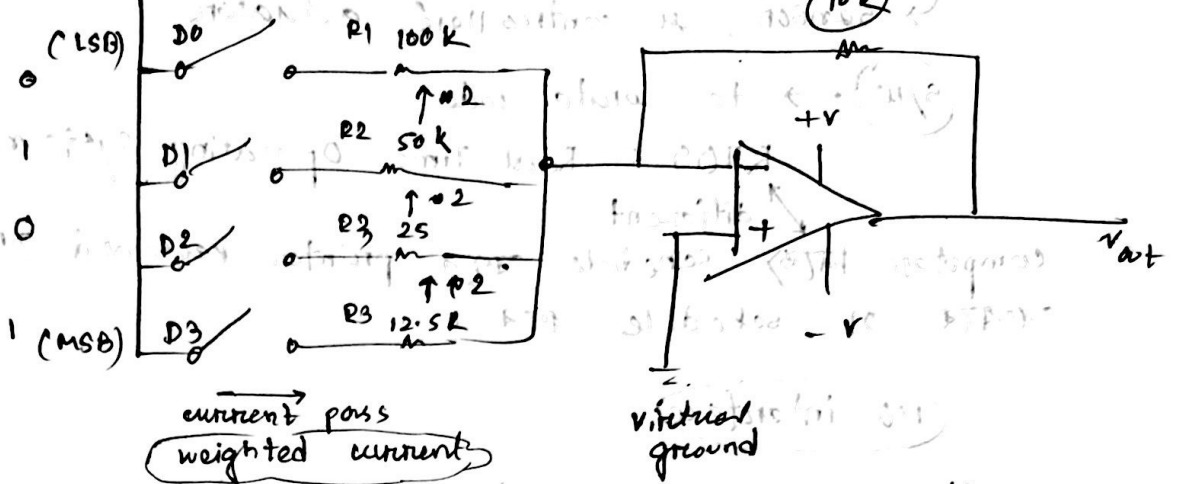
$D_3 D_2 D_1 D_0$

(4 bit) $\rightarrow$ 4 switch

Ref. V (V)

(LSB)
 $(2^{n-1})R$
 $\vdots$
 $2^2 R$
 $2R$
 (MSB) R

weighted resistance



$$V_{out} = -R_f \left(\frac{D_0(-V)}{R_1} + \frac{D_1(-V)}{R_2} + \frac{D_2(-V)}{R_3} + \frac{D_3(-V)}{R_4} \right)$$

(inverting point 2 connect)

0000 $\rightarrow$ 0V

0001 $\rightarrow$ $V = 5V$
 $R_1 = 100k$ $R_f = 10k$

$$V_{out} = \frac{-10(-5)}{100} = 0.5V$$

0010 $\rightarrow$ 1V

0011 $\rightarrow$ 1.5V

0100 $\rightarrow$ 2.0V

1111 $\rightarrow$ 5V

max $\rightarrow$ 5V
 2^4 comb.

If $R_f = R_1$
 $\rightarrow$ 1 step $\rightarrow$
 $\frac{V_{ref}}{2^n}$

($R_f = 10k$
 $R_1 = 100k$)

Resolution $\rightarrow$

4 bit $\rightarrow$ 1 part in 16

$$n \text{ bit} \rightarrow \left(\frac{1}{2^n}\right) \text{ or } \left(\frac{1}{2^n} \times 100\%\right) = \text{resolution}$$

Full scale o/p voltage $\rightarrow$

$V = \text{Ref. V} = \text{Full scale voltage}$ (2374 डिग्री)

$$\text{actual FS. o/p voltage} \rightarrow \max(V) - \frac{V}{2^n} \quad (1 \text{ step})$$

$$5 \rightarrow 1 \text{ step} = 0.5 \\ (5 - 0.5 = 4.5 \text{ V})$$

let, 12 bit, 10V converter

$$1 \text{ step} = \frac{10}{2^{12}}$$

$$\text{actual} \rightarrow \left(10 - \frac{10}{2^{12}}\right) \text{ V.}$$

$$\text{Ex : } 5 - \frac{5}{2^n} = 4.6875$$

$$\Rightarrow n = 4$$

Accuracy :

let, 0010 $\rightarrow$ calc $\rightarrow$ 1.0 V

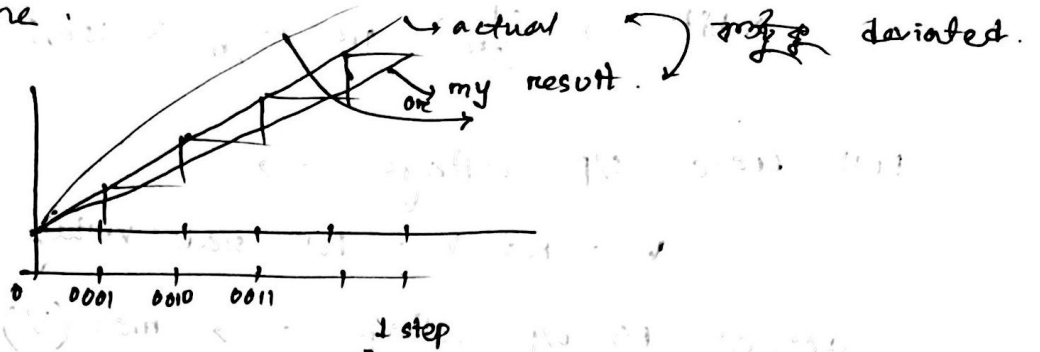
$$\text{Ex : } 0.2\% \text{ accuracy} \rightarrow 10 \times \frac{0.2}{100} = 20 \text{ mV}$$

आजान्न रमा 9.5 V

but, 9.5 V $\pm$ 20 mV correct वैज्ञानिक

Linearity :

how much o/p ramp deviates from straight line.



→ Linear ($\pm \frac{1}{2}$ LSB এর মধ্যে থাকলে)

→ Non-linear

let, $\pm \frac{1}{2} \text{ LSB} = \pm \frac{1}{2} \times 0.5 = \pm 0.25$

0011 → 1.5 আসার কথা

1.39 আসলে → Linear system.

settling time →

updown করতে যতক্ষণ

error range → ± 0.002

0010 → 1 ✓

0.005

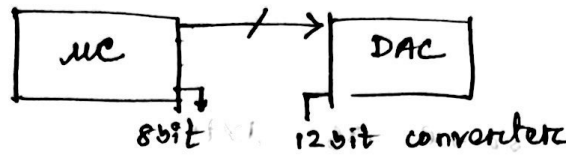
0.95 (actual value)

0.99 → error range এর মধ্যে চলে আসলে,

time that takes → settling time

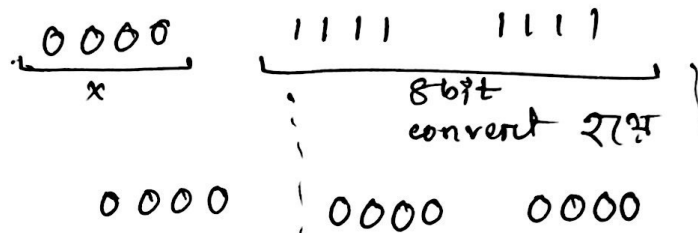
digital o/p $\rightarrow$ μ controller

D/A c to convert digital o/p.



1st 4, 4 bit latch $\rightarrow$ 12 bit pass.
8 bit directly pass

Latch use na jhalan



0000 0000 0000 $\rightarrow$ but
value 0
consider
value which
will generate
unwanted
result.

then, we write upper 4 bits

